Shloke Binani

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EDUCATION

Vellore Institute of Technology, Vellore, India — 2023 to 2027
Bachelor of Technology in Electronics and Communication Engineering | **CGPA: 8.68 / 10.0**

Haryana Vidyamandir School, Kolkata, India — 2022 to 2023
Higher Secondary Examination (CBSE, Class XII) | **Score: 80%**

Calcutta Boys' School, Kolkata, India — 2020 to 2021
Indian Certificate of Secondary Education (ICSE, Class X) | **Score: 93.2%**

TECHNICAL SKILLS

- **Languages & Core Concepts**: Java (DSA), Python, C++, C, Operating Systems (OS), Computer Networks (CN), DBMS
- **AI & Machine Learning**: Generative AI, RAG (Retrieval-Augmented Generation), QCNN, CNN, RNN, ANN, Random Forest, PCA, Clustering (K-Means, DBSCAN), Hypothesis Testing, Model Evaluation, LLMs, AI-Assisted Coding
- **Backend & Deployment**: FastAPI, Docker, Microservices Architecture, REST APIs, GraphQL, Hasura, CI/CD, Git, Linux
- **Data Engineering & Databases**: Python (Pandas), Databricks, ETL Pipelines, SQL, PostgreSQL, MySQL, MongoDB, Power BI, Tableau, MATLAB

EXPERIENCE

Kofuku Technologies Pvt. Ltd., On-Site — May 2025 to July 2025
Software Engineer Intern | AI Backend Systems & Automated Pipelines

- Cut deployment overhead by **35%** by containerizing **FastAPI** microservices for voice-AI scheduling via **Docker**
- Reduced manual collection by **70%** by engineering a 3-stage **n8n ETL data pipeline** for influencer ingestion
- Resolved **JWT** authentication and authorization faults across services by enforcing secure coding practices
- Improved API test coverage by **35%** by validating services across **Hasura**, **MySQL**, **MongoDB**, and **Postman**
- Lifted survey completion by **40%** across **150+ users** by refactoring data workflows and cutting defects by **25%**

ACADEMIC PROJECTS

MS-HQNet: Multi-Stream Hybrid Quantum Framework | *Python, Machine Learning, Signal Processing*

- Developed a hybrid quantum-classical neural network combining a 6-qubit **QCNN** and an **MLP** bypass to classify multi-channel EMG signals for eye movement prediction
- Engineered high-throughput signal preprocessing and feature extraction **data pipelines**, outperforming baseline models to achieve **99.17% accuracy**

Customer Attrition Predictive Modeling | *Python, SQL, Scikit-Learn, Random Forest* — GitHub

- Built data processing pipelines across **7K+ customer records** in **SQL** and **Python** to isolate behavioral churn drivers
- Trained and tuned a **Random Forest** classifier (**80% accuracy**), applying statistical hypothesis testing and feature importance scoring to model retention intervention scenarios

Retail Analytics & Customer Segmentation | *Python, SQL, Power BI, PCA, Clustering* — GitHub

- Developed an unsupervised machine learning pipeline on transactional data, implementing **PCA** dimensionality reduction and hybrid **DBSCAN/K-Means** clustering
- Deployed an interactive **Power BI** dashboard and implemented an **A/B testing** statistical framework in **Python** to evaluate cohort conversion metrics

LEADERSHIP

Society of Automotive Engineers (SAE-VIT) | *Marketing, Design & PR Head* — May 2024 to May 2026

- Directed outreach campaigns reaching **1,000+ students**, securing INR 1 Lakh in chapter sponsorships
- Served as operations manager for SAE AirShow 25 at graVITas 25, coordinating **40+ team members** to drive a **30 to 45%** increase in event turnout
- Managed event logistics, technical equipment scheduling, and vendor coordination across **12 major university events**, ensuring 100% on-time execution

ACADEMIC ACHIEVEMENTS & CERTIFICATIONS

- Solved **250+ problems** on LeetCode & GeeksforGeeks (primarily in Java), covering DSA and SQL
- Awarded the **Elite Category Chapter Award** for SAE-VIT for excellence in operational management, 2025 to 2026
- Awarded the **Leadership Certificate** for exceptional contribution and management by Vellore Institute of Technology
- Won **Runner-Up** position at Central Hack for leveraging Web3 Crisis Protocols
- Won **Runner-Up** position at Providence Leap Ideathon for AI-Driven Healthcare Benefit Orchestration

Certifications: Pandas, Kaggle; Statistics OnRamp (2024) & MATLAB OnRamp (2024), The MathWorks, Inc.